of any one or more beads denotes deletion type of chromosomal aberration).

3. For Duplication – Give yellow colour beads with addition of one or more marked alphabets A to H (The repetition of one or more beads denotes duplication type of chromosomal aberration).
4. For Inversion – Give yellow colour beads which marked alphabets from A to H as in normal chromosome. (There is no addition or deletion of beads (A to H) given, so the students can construct the inverted segment of the chromosome using the given beads).

Based on the type of beads given the student has to identify and construct the relevant chromosomal aberration.

## 13 A. Chromosomal Aberration – Deletion

**Reasons:**

1. The deletion of the chromosomal segement A and D. (Refer figure 13a)
2. When there is a loss of a segment of the genetic material in a chromosome it is called deletion.

**Significance:**

Most of the deletions lead to death of an organism.

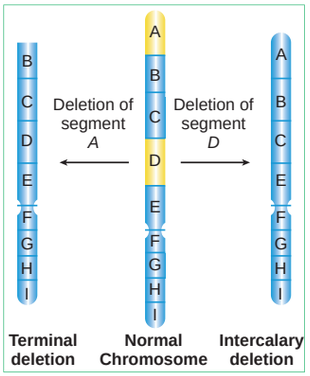



**Figure: 13 a:** Deletion

## 13 B. Chromosomal Aberration - Duplication

**Reasons:**

1. When a segment of a chromosome is present more than once in a chromosome, then it is called duplication (Tandem duplication)
2. The order of the genes in a chromosome is A, B, C, D, E, F, G, H and I. Due to aberration, the genes B and C are duplicated and the sequence of genes becomes A, B, C, B, C, D, E, F, G, H and I. (Refer figure 13b)

**Significance:**

Some duplications are useful in the evolution of the organism.

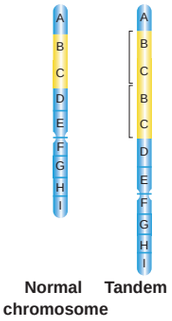



**Figure 13 b:** Duplication

## 13 C. Chromosomal Aberration - Inversion

**Problem:**

Given below is the representation of a kind of chromosomal aberration. Identify it giving reasons for your identification. Also mentions its significance.

**Identification:**

The given genetic problem is identified as inversion type of chromosomal aberration.

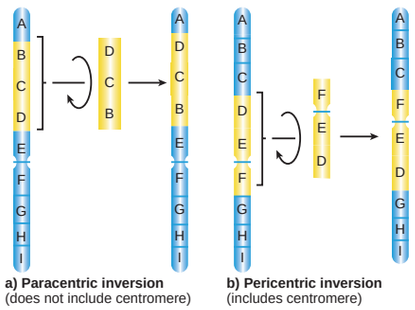



**Figure: 13 c:** Inversion

**Reasons:**

1. When the order of genes in a chromosomal segment is reversed due to rotation by an angle of 180°, it is called inversion.
2. The order of genes in a chromosome is A, B, C, D, E, F, G, H and I. Due to aberration, the sequence of genes become A, D, C, B, E, F, G, H and I (Refer figure 13c)



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